🧠 EEG Details to Notice

1. Room & Environmental Setup

- **Electrical noise minimization**: Turn off unnecessary electronic devices.
- **Lighting**: Avoid fluorescent lights that can introduce 50/60 Hz noise.
- **Temperature**: Maintain a comfortable room temperature for participant comfort.
- **Chair setup**: Use a comfortable, preferably reclining chair for participants.
- **Emergency protocols**: Ensure all safety procedures and first-aid supplies are accessible.

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2. System & Hardware Check

- **EEG amplifier**:
    - Properly connected. (Make sure it is clicked locked on the base.)
    - Connected to the acquisition computer.
    - Firmware/software versions up to date.
- **Electrode cap (if using one)**:
    - Correct size for participants.
    - Clean and intact.
    - No broken wires or loose connectors.
- **Electrodes**:
    - Cleaned with water.
    - Confirm correct impedance range (<5–10 kΩ generally).
- **Cables and connectors**:
    - Gently plugged in the connector (*make sure it is clicked locked)

    - Gently connect the connector with the cap.
    - Avoid wires tangling.
- **Gel/paste & syringe**:
    - Use conductive gel (check expiry).
    - Avoid over-application to prevent bridging between electrodes.
- **Electrode placement system (10-20)**:
    - Mark key landmarks (nasion, inion, preauricular points) for accurate positioning. (Make sure the CZ channel is in the center.)
- **Impedance checking device**:
    - Calibrated and functioning properly.

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3. Computer & Software Setup

- **Acquisition software**:
    - Correct protocol loaded.
    - Channel configuration matches the cap setup.
- **Event markers/synchronization system**:
    - Confirm that stimulus presentation software (e.g., Psychtoolbox, E-Prime, PsychoPy) is synchronized.
    - Parallel port triggers are configured and tested.
- **Backup system**:
    - Data saving location confirmed.
    - Backup drive ready if needed.

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4. Participant Preparation

- **Hair preparation**:
    - Clean scalp using alcohol swabs or abrasive gel.
    - Tie up long hair or secure it with clips (non-metallic) or hair tie in a low position.
- **Skin preparation**:
    - Lightly apply skin scrubs gel under electrodes of horizontal & vertical eye movements channels and mastoids to reduce impedance.
- **Comfort**:
    - Adjust the cap snugly but not too tight.
    - Provide instructions and answer questions clearly.
- **Contraindications**:
    - Screen for scalp injuries, skin allergies, or implants that could interfere with recording.

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5. Final Assembly & Pre-recording Checks

- **Cap secured and aligned** (Cz at vertex).
- **All electrodes connected correctly**.
- **Ground and reference electrodes**:
    - Secure placement and stable contact.
- **Check impedance levels (<5–10 kΩ at best).**
- **Do a test recording**:
    - Confirm low noise.
    - Blink and movement artifacts present in expected channels.
- **Check for electrode bridging**.
- **Check synchronization signal between stimulus & EEG**.

- Borrow alternative electrodes if needed when channels were malfunctioning.

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6. Documentation

- Record:
    - Cap size, electrode gel used, impedance values.
    - Time of setup and start of recording.
    - Any issues encountered.
    - Participant ID and informed consent.

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7. Space Restoration & EEG Cleaning

- **Cap and electrodes**:
    - Remove and rinse with cold water immediately after use.
    - Gently brush away gel residues using cotton swabs.
    - Inspect for damage or wear (cracks, broken wires, bent pins).
    - Dry thoroughly and store in a cool, dry place.
- **Consumables**:
    - Properly dispose of used swabs, tissues, and 3M tapes.
- **Workstation cleanup**:
    - Wipe down chair, table, and any reused equipment surfaces.
    - Reorganize cables and return unused materials to storage.
- **Electronics**:
    - Turn off the software before unplugging the amplifier and computer.
- **Room reset**:

    - Restore room to baseline setup for the next participant or team.

- **Incident reporting**:
    - Note any equipment issues, maintenance needs, or protocol deviations.